

Hamro Hospital

Hospital Deployment Guide

1. Deployment Overview

Hamro Hospital can run locally for small hospitals or on cloud/LAN servers for larger hospitals. SQLite is for development; PostgreSQL is recommended for production.

2. Required Hardware

Users	Recommended server
500	8 CPU cores, 16-32GB RAM, NVMe SSD, PostgreSQL on same/LAN server
1,000	16 cores, 32-64GB RAM, dedicated PostgreSQL, Redis
5,000	App servers behind load balancer, 32+ cores aggregate, 128GB RAM DB, read replicas
10,000	HA architecture, multiple app servers, managed PostgreSQL cluster, Redis cluster, CDN/backups

3. Network and Devices

Use a stable LAN for hospital counters. Barcode scanners can work as keyboard-wedge scanners. Receipt printers should be installed on counter machines. Fingerprint machines should export CSV/API data to the device punch endpoint.

4. PostgreSQL Setup

Set `DJANGO_DB_ENGINE=postgres` and configure `POSTGRES_DB`, `POSTGRES_USER`, `POSTGRES_PASSWORD`, `POSTGRES_HOST`, `POSTGRES_PORT`. Run `python manage.py migrate`.

5. Redis, WebSockets and Celery

Set REDIS_URL and run ASGI with Daphne/Uvicorn. Start Celery worker and optional beat for scheduled jobs.

6. eSewa Setup

Set ESEWA_SANDBOX=False, ESEWA_MERCHANT_CODE and ESEWA_SECRET_KEY. Test callback URLs before live use.

7. Backup and Recovery

Schedule database dumps, media backups and offsite/cloud backups. Test restore procedure regularly.

8. User Setup

Create Super Admin, departments, department heads, staff accounts, doctors, finance users, counters and service catalogues.

9. Daily Operation

Registration creates patients/OPD, doctors send referrals/orders, departments generate pending bills, cashier collects, departments process paid services, finance reconciles, attendance/payroll tracked.

10. Approximate Costs

Item	Approx Nepal estimate
Local server	NPR 150,000 - 800,000+
Cloud VPS	NPR 3,000 - 50,000/month depending scale
Domain	NPR 1,500 - 4,000/year
SSL	Free LetsEncrypt or paid NPR 5,000+/year
SMS/OTP	Provider dependent per SMS
Maintenance	NPR 20,000 - 200,000+/-month depending SLA